Tuberculosis & Pneumonia Extract

Compiled from the Nepalese National Formulary (NNF) 2018, 3rd Edition

This document brings together the pages of the NNF 2018 that deal specifically with tuberculosis and pneumonia — the dedicated anti-tubercular drug monographs, the antibacterial monographs whose stated indications include pneumonia, related diagnostic/immunisation content, key drug-interaction entries, and the relevant essential-medicines listing. Original page layout, dosages and page numbers are reproduced unchanged from the source document.

Contents

- Rational drug combinations — anti-tubercular fixed-dose combinations (p. 6)
- Chapter 10 overview — Drugs used in Infections (index of sub-sections) (p. 237)
- Cephalosporins — Cefaclor, Cefadroxil, Cefazolin, Cefepime, Cefixime, Cefoperazone, Cefotaxime, Cefpodoxime, Ceftazidime, Ceftriaxone, Cefuroxime (pp. 242–245)
- Penicillins — Amoxicillin, Ampicillin, Bacampicillin, Benzylpenicillin, Carbenicillin, Cloxacillin, Flucloxacillin, Piperacillin, Piperacillin/Tazobactam (pp. 246–252)
- Lincosamide, Oxazolidinone & Macrolides — Clindamycin, Linezolid, Azithromycin, Clarithromycin, Erythromycin (pp. 254–257)
- Quinolones & Fluoroquinolones — Ciprofloxacin, Levofloxacin, Moxifloxacin, Nalidixic acid, Norfloxacin, Ofloxacin (pp. 257–260)
- Pentamidine — visceral leishmaniasis and Pneumocystis pneumonia (pp. 270–271)
- Dapsone — includes prophylaxis/treatment of Pneumocystis pneumonia (p. 273)
- Anti-tubercular drugs, first line — Ethambutol, Isoniazid, Pyrazinamide, Rifampin, Streptomycin (pp. 277–279)
- Anti-tubercular drugs, second line — Bedaquiline, Capreomycin, Cycloserine, Ethionamide, Kanamycin (pp. 280–281)
- Tuberculin, Purified Protein Derivative (PPD) — Mantoux skin test (p. 368)
- National immunisation schedule — BCG (tuberculosis) and PCV (pneumococcal pneumonia) (p. 378)
- Appendix 1, Drug Interactions — dedicated Rifampicin and Streptomycin interaction entries (pp. 538–541)
- Appendix 2, National List of Essential Medicines 2016 — anti-tubercular medicines, main and complementary lists, incl. second-line MDR-TB medicines (p. 556)

Note: page numbers above refer to the printed page numbers of the original NNF 2018. This is a curated extract, not an official Government of Nepal publication; for the complete formulary and any updates, refer to the full NNF 2018 document.

3. Drug combinations - how rational are they?

Fixed-dose combination products are acceptable only when the dose of each ingredient meets the requirement of a defined population group and when the combination provides a proven advantage over single compound administered separately in therapeutic effect, safety or compliance.

Fixed-dose combinations are only acceptable if the following criteria are met:

1. Clinical documentation justifies the concomitant use of more than one drug.
2. The therapeutic effect is greater than the sum of the effect of each.
3. The cost of the combination product is less compared to the sum of the individual products.
4. Compliance is improved.
5. Sufficient drug ratios are provided to allow dosage adjustment satisfactory for the majority of the population.

Examples of useful drug combinations in Nepal's National List of Essential Medicines, 2016 are as follows:

- Amoxicillin + clavulanic acid (anti-bacterial)
- Artemether + lumefantrine (antimalarial)
- Benzoic acid + salicylic acid (for external use)
- Ethambutol + isoniazid (anti-tubercular)
- Ethambutol + rifampicin + isoniazid (anti-tubercular)
- Ethambutol + rifampicin + isoniazid + pyrazinamide (anti-tubercular)
- Ichthammol + glycerine (ear drop)
- Isoniazid + rifampicin (anti-tubercular)
- Isoniazid + rifampicin + pyrazinamide (anti-tubercular)
- Ethinylestradiol+ levonorgestrel (oral contraceptive)
- Ethinylestradiol + norethisterone (oral contraceptive)
- Ferrous sulfate + folic acid (anti-anaemic)
- Lopinavir + ritonavir (anti-viral)
- Levodopa + carbidopa (Parkinson's disease)
- Lidocaine + epinephrine (local anaesthetic)
- Mifepristone + misoprostol
- Saquinavir + ritonavir (anti-viral)
- Sulfadoxine + pyrimethamine (anti-malarial)
- Sulfamethoxazole + trimethoprim (anti-bacterial)
- Zidovudine + Lamivudine + Nevirapine (anti-retrovirals)

4. Prescription writing

A prescription is an instruction from a prescriber to a dispenser. The prescriber is not always a doctor but can be a paramedical worker, such as health assistant and community health workers (with restrictions). The dispenser is not always a pharmacist, but can be a pharmacy technician, an assistant, or a layperson. Thus, it is essential that following guidelines, which are not exclusive, for prescription writing be followed:

1. Prescriptions should be written legibly in ink and should be dated. The local language is preferred.
2. The prescriber's name, address and telephone number so that s/he can be

Section II - Chapter 10
Drugs used in Infections

10.1 Anti-amoebic and Anti-giardial drugs
Diloxanide furoate
Metronidazole
Secnidazole
Tinidazole

10.2 Antibacterial (other than antitubercular and antileprotic drugs)
10.2.1 Cell wall synthesis inhibitors
10.2.1.1 Cephalosporins
Cefaclor
Cefadroxil
Cefalexin
Cefazolin
Cefepime
Cefixime
Cefoperazone
Cefotaxime
Cefpodoxime
Ceftazidime
Ceftriaxone
Cefuroxime
10.2.1.2 Penicillins
Amoxicillin
Amoxicillin/clavulanate
Ampicillin
Ampicillin/sulbactam
Bacampicillin
Benzyl penicillin (penicillin G)
Carbenicillin
Cloxacillin
Flucloxacillin
Phenoxy methyl penicillin (penicillin V)
Piperacillin
Piperacillin/tazobactam
10.2.2 Protein synthesis inhibitors
10.2.2.1 Aminoglycosides
Amikacin
Gentamicin
Neomycin
10.2.2.2 Chloramphenicol
10.2.2.3 Lincosamide derivative
Clindamycin
10.2.2.4 Oxazolidinones
Linezolid

Drug and food interaction: Don't use within 2 weeks of use of disulfiram. Cimetidine and ketoconazole increases effects of tinidazole. Phenobarbital, rifampin, phenytoin, cholestyramine, oxytetracycline decreases effects. Increases action of anti-coagulants, cyclosporine, tacrolimus, hydantoins, and lithium.

10.2 Anti-bacterial (other than anti-tubercular and anti-leprotic drugs)

10.2.1 Cell wall synthesis inhibitors

10.2.1.1 Cephalosporins

CEFACLOR

Dosage form and strengths: *Capsule:* 250 mg, 500 mg; *Oral suspension:* 125 mg/5 ml.

Indications: Susceptible infections due to sensitive gram-positive and gram-negative bacteria, pneumonia, lower UTI.

Contraindications/Precautions: Cephalosporins have cross-allergenicity with penicillins (about 10% cases) so it should be avoided in patients who have had hypersensitivity reaction to penicillins. Prolonged use of cephalosporins may result in the overgrowth of non-susceptible organisms especially *Pseudomonas, Enterococci* or *Candida*.

Dosage schedule: 250 mg every 8 hours, doubled for severe infection; maximum 4 g daily. Child over one month: 20 mg per kg daily in 3 divided doses, doubled for severe infections, maximum 1 g daily.

Adverse effects: Hypersensitivity reactions including urticaria, pruritus, rash, fever, joint pain and exfoliative dermatitis. Anaphylaxis occurs rarely. Nausea, vomiting, diarrhoea, positive coomb's test, elevated liver enzymes levels and rarely thrombocytopenia or neutropenia.

CEFADROXIL

Dosage form and strengths: *Capsule:* 500 mg, 1000 mg; *Tablet:* 125 mg, 250 mg; *Oral suspension:* 125 mg/5 ml, and 250 mg/ml.

Indication: Skin and soft tissue infection, uncomplicated UTI, pharyngitis, tonsillitis, endocarditis, renal impairment.

Contraindications/Precautions Hypersensitivity. See under cefaclor.

Dosage schedule: 250 mg oral every 6 hours or 500 mg every 8-12 hours increased to 1-1.5 g every 6-8 hours for severe infections; child: 30 mg/kg daily in divided doses, doubled for severe infections, maximum 100 mg/kg daily.

• *Prophylaxis of recurrent urinary tract infection in adult:* 125 mg at night.

Adverse effect: See under cefaclor.

CEFALEXIN (Cephalexin)

Dosage form and strengths: *Capsule:* 250 mg, 500 mg; *Tablet:* 125 mg; *Oral suspension:* 125 mg/5 ml, 250 mg/5 ml.

Indications: Susceptible infections due to sensitive gram-positive and gram-negative bacteria, pneumonia, lower UTI.

Contraindications/Precautions: See under cefaclor.

Dosage schedule: 250 mg oral every 8 hours, doubled for severe infection; maximum 4 g daily. Child>1 month: 25-50 mg per kg daily in 3 divided doses, doubled for severe infections, maximum 1 g daily.

- *Prophylaxis of recurrent urinary tract infection:* adult: 125 mg at night; child: 10 mg/kg at night.

Adverse effects: See under cefaclor.

Drug and food interaction: Increases effects of anticoagulants (use cautiously); increases toxicity of aminoglycosides, loop diuretics, probenecid. Decreases effects of oral contraceptives, use alternative methods of contraception.

CEFAZOLIN

Dosage form and strengths: *Powder for injection:* IV/IM: 1 g, 2 g, 500 mg.

Indication: Surgical prophylaxis, susceptible infections due to sensitive gram-positive and gram-negative bacteria, pneumonia, lower UTI.

Contraindications/Precaution: Dose must be adjusted in severe renal insufficiency with caution in patient with seizure.

Dosage schedule: *Intramuscular injection or intravenous injection or infusion:* 0.5-1 g every 6-12 hours; Child: 25-50 mg/kg daily in divided doses, increased to 100 mg/kg daily in severe infection.

Adverse effects: See under cefaclor.

CEFEPIME

Dosage form and strengths: *Vial*: 1 g and 2 g.

Indications: See under cefaclor.

Contraindications/Precautions: See under cefaclor.

Dosage schedule: Intravenous injection, 2 g every 12 hours. Child:100-150 mg/kg/day

Adverse effects: See under cefaclor.

CEFIXIME

Dosage form and strengths: *Capsule/tablet*: 100 mg, 200 mg, 400 mg**;** *Syrup:* 100 mg/5 ml

Indications: See under cefaclor.

Dosage schedule:

Adult and child over 10 years: 200-400 mg daily in 1-2 divided doses, child: 8 mg/kg in 1-2 divided doses.

Gonorrhoea: 400 mg as a single dose.

Adverse effects: See under cefaclor.

CEFOPERAZONE

Dosage form and strengths: *Vial:* 1 g, 2 g.

Indications: Pelvic infections, urinary-tract infections, bone and joint infections.

Contraindications/Precautions: It may produce disulfiram like reaction with alcohol.

Dosage schedule: By intramuscular or intravenous injection: 1-2 g (base) every 12 hours.

Adverse effects: See under cefaclor.

CEFOTAXIME

Dosage form and strengths: *Powder for injection:* 250 mg, 500 mg and 1 g.
Indications: Upper respiratory tract infection, uncomplicated UTI, gonorrhea.
Dosage schedule: By intramuscular or intravenous injection 1 g every 12 hours. Severe infections, 2 g every 6 hours, exceptionally, for life threatening infection due to less sensitive organism. Child: 100-150 mg/kg/day in 3-4 divided doses (max. 200 mg/kg/day).
Adverse effects and precautions: See under cefaclor.

CEFPODOXIME

Dosage form and strengths: *Oral suspension:* 50 mg/ml, 100 mg/ml; *Tablet:* 100 mg, 200 mg
Indications: Upper and lower respiratory tract infections, uncomplicated urinary tract infections, skin and soft tissue infections, uncomplicated gonorrhea.
Contraindication and Precautions: Documented hypersensitivity, renal impairment
Dosage schedule:

- *Upper respiratory-tract infections:* 100 mg twice daily (200 mg twice daily in sinusitis) for 10 days; child: 8 mg /kg/day in 2 divided doses.
- *Uncomplicated urinary-tract infections:* 100 mg twice daily (200 mg twice daily in uncomplicated upper urinary-tract infections); Child: 8 mg/kg/day in 2 divided doses.
- *Uncomplicated gonorrhea:* 200 mg as a single dose.
- *Skin infection:* 400 mg oral BD 7-14 days.

Adverse effects: See under cefaclor.

CEFTAZIDIME

Dosage form and strengths: *Powder for injection:* 250 mg, 500 mg, 1 g.
Indications: See under cefpodoxime.
Dosage schedule: By deep intramuscular injection or intravenous injection or infusion: 1 g every 8 hours; Severe case: 2 g every 12 hours; child up to 2 months: 25-60 mg/kg daily in 2 divided doses; Over 2 months: 30 -100 mg/kg daily in 2-3 divided doses, up to 150 mg/kg daily (maximum 6 g daily) in immunocompromised or meningitis.

- *Pseudomonal lung infection in cystic fibrosis:* Adult with normal renal function: 100-150 mg/kg daily in 3 divided doses; Child: upto 150 mg/kg daily (maximum 6 g daily) in 3 divided doses; preferably i.v.
- *Urinary tract and less severe infections:* 0.5-1 g every 12 hours.
- *Surgical prophylaxis, prostatic surgery:* 1 g at induction of anesthesia repeated if necessary when catheter is removed.
- *Febrile neutropenia:* IV/IM: 2 g every 8hrs.

Adverse effects and precautions: See under cefaclor.

CEFTRIAXONE

Dosage form and strengths: *Injection:* IV/IM: 250 mg, 500 mg, 1000 g, 2 g
Indications: See under cefpodoxime, epiglottitis.
Contraindications and Precautions: Hypersensitivity, hyperbilirubinemic neonates particularly those who are premature; neonates <28 days if they

receive calcium containing iv products. Risk of fatal calcium-ceftriaxone precipitant formation in lungs and kidneys of term and preterm neonates. May increase INR , especially in nutritionally deficient patients.

Dosage schedule: By deep intramuscular injection, or by intravenous injection over 2-4 minutes, or by intravenous infusion, 1 g daily as single dose; 2-4 g daily as a single dose in severe infection.

- *Surgical prophylaxis:* 1 g 30 min before procedure.
- *Early syphilis:* IM 500 mg daily for 10 days. Susceptible infection due to sensitive gram positive and gram negative bacteria IM/IV 1 g daily, in severe form:-2-4 g daily.
- *Meningitis:* Child 1 month – 11 yrs.: IV 50-75 mg/kg single dose or in 2 divided doses (max. 100 mg/kg/day). Neonates: use with cautions.
- *Epiglottitis:* oral, intravenous: 1 g twice in a day, then step down to 200 mg BD cefixime + oral steroid stat (hydrocortisone).

Adverse effects: CNS: headache, dizziness, weakness, fever, seizures**.** GI: nausea, vomiting, deranged LFT, abdominal pain**.** Hematological: leukopenia, thrombocytopenia, agranulocytosis, rash, utricaria, anaphylaxis, toxic epidermal necrolysis.

Food and drug interaction: Increase cyclosporine level. Increase bleeding-anticoagulants, thrombolytic, NSAIDS. Increase toxicity-aminoglycosides, probenecid. Decrease absorption-iron rich cereals.

CEFUROXIME

Dosage form and strengths: *Oral suspension:* 125 mg/5 ml, 250 mg/5 ml; *Tablet* 250 mg, 500 mg; *Powder for injection:* 750 mg, 1.5 g, 7.5 g, 75 g, 225 g.

Indications: Surgical prophylaxis. It is more active against *Haemophilus influenzae* and *Neisseria gonorrhoeae,* tonsillitis and pharyngitis.

Contraindications/Precautions: If diabetic, use blood glucose test. To complete the course of dose.

Dosage schedule: Oral (as cefuroxime axetil), 250 mg twice daily in most infections including mild to moderate respiratory tract infections.

- *Urinary-tract infections:* 125 mg twice daily, doubled in pyelonephritis.
- *Gonorrhoea:* 1 g as a single dose. Child: IV 50-100 mg/kg/day and Oral 20-40 mg/kg/day. Adult: 750 mg every 6-8 hrs, increase if necessary up to 1.5 g every 6-8 hrs, increased dose for severe infection.
- *Surgical prophylaxis:* 1.5 g by intravenous injection at induction; may be supplemented with 750 mg intramuscularly or intravenous every 8 hrs if required for up to 3 doses.
- *Tonsillitis and pharyngitis:* adult, 250 mg oral once daily for 4days, child: 10 mg/kg PO BID for 4-10 days.

Adverse effects: See under cefaclor.

Drug and food interaction: Increase effect/toxicity: aminoglycoside, furosemide, probenecid. Increase bleeding risk: anticoagulants, thrombolytics, and antiplatelets. Decrease oral contraceptive if possible use alternative one. Decrease absorption of cephalosporin: antacid. Decrease effect of cephalosporin: H_2 blockers

10.2.1.2 Penicillins

AMOXICILLIN

Dosage form and strengths: *Capsule*: 250 mg and 500 mg; *Dispersible tablets*: 125 mg and 250 mg, *Drops:* 100 mg/1ml, *Injections:* 100 mg, 250 mg, 500 mg; *Oral suspension:* 125 mg/5 ml, 250 mg/5 ml.

Indications: Upper respiratory tracts infections (by *S. pneumoniae, S. pyogens, H. influenzae*), otitis media, sinusitis, pneumonia (by *H. influenzae*), pharyngitis, skin and soft tissue infections, genitourinary tract infection, *H. pylori* infection, diphtheria.

Contraindications/Precautions: Documented hypersensitivity to penicillins, cephalosporins, imipenem. Clostridium difficile-associated diarrhoea (CDAD), infectious mononucleosis, suspected bacterial infection, superinfections with bacterial or fungal pathogens, allergy to cephalosporins, carbapenems. Take medication without regard to meals.

Dosage schedule:

- *Ear, nose & throat infections:* mild to moderate: 500 mg oral q12hr or 250 mg oral q8hr for 10-14 days; severe: 875 mg oral q12hr or 500 mg oral q8hr for 10-14 days.
- *Tonsillitis/pharyngitis:* 875 mg oral qday for 10 days, taken within 1 hour after finishing a meal. Lower respiratory tract infections: 875 mg oral q12hr or 500 mg oral q8hr for 10-14 days.
- *Skin & skin structure infections:* mild to moderate: 500 mg oral q12hr or 250 mg oral q8hr; severe: 875 mg oral q12hr or 500 mg oral q8hr.
- *Genitourinary tract infections:* mild to moderate: 500 mg oral q12hr or 250 mg oral q8hr; severe: 875 mg oral q12hr or 500 mg oral q8hr.
- *Helicobacter pylori infection:* Triple therapy: 1 g PO q12hr for 14 days with lansoprazole (30 mg) and clarithromycin (500 mg); dual therapy: 1 g PO q8hr for 14 days with lansoprazole (30 mg) in patients intolerant of, or resistant to, clarithromycin.
- *Diphtheria:* 500 mg TDS for 2 weeks (along with diptheria antitoxin up to 1,00,000 U IV)

Adverse effects: Anaphylaxis, anaemia, AST/ALT elevation, mucocutaneous candidiasis, diarrhoea, headache, nausea, vomiting, rash, pseudomembranous colitis, serum sickness-like reactions

Drug and food interactions: BCG vaccine live, cholera vaccine, demeclocycline, doxycycline, minocycline, mycophenolate, tetracycline, typhoid vaccine live, acyclovir, allopurinol, aspirin, bendroflumethiazide, chlorothiazide, cyclopenthiazide, daclizumab, dienogest, estradiolvalerate, estradiol, ethinylestradiol, hydrochlorothiazide, mestranol, methotrexate, metolazone, warfarin, amiloride, azithromycin

AMOXICILLIN with CLAVULANATE

Dosage form and strengths: *Oral suspension:* (125 mg/31.25 mg)/5 ml; *Tablet*: 500 mg (amoxicillin) with 125 mg (clavulanic acid); Injection, 1 g (amoxicillin) with 200 mg (clavulanic acid)

Indications: Lower respiratory tract infection (LRTI), chronic obstructive pulmonary disease, acute bacterial sinusitis, acute mastoiditis, animal and human bite wounds, erysipelas, pyelonephritis, skin abscess, tonsillitis and

pharyngitis.

Contraindications/Precautions: Allergy to penicillins, previous history of cholestatic jaundice/hepatic dysfunction associated with amoxicillin/clavulanate. Allergy to cephalosporins, carbapenems. Use caution in hepatic impairment. Hepatic dysfunction (rare in adult) is more common in elderly and/or males and prolonged therapy may increase risk. Take with meals to avoid GI upset, take suspension at start of meal to enhance absorption. In case of dysphagia: may substitute 250 mg/5 ml suspension for 500/125 mg tablet; may substitute 200 mg/5 ml or 400 mg/5 ml suspension for 875/125 mg tablet.

Dosage schedule:

- *LRTI:* Beta lactamase producing strains of Haemophilus influenza and Moraxella catarrhalis. Mild to moderate: 500/125 mg oral q12hr or 250/125 mg oral q8hr for 10 days. Severe: 875/125 mg oral q12hr or 500/125 mg oral q8hr for 7-10 days. COPD: 500 mg oral q8hr.
- *Acute bacterial sinusitis due to beta lactamase producing strains of H. influenza & M. catarrhalis:* 2000 mg (2 extended release Tablets) PO q12hr for 10 days.
- *Acute mastoiditis:* 1.2 g IV q8hrly then step down to oral 625 mg q8hrly for total of 10 -14 days.
- Animal/human bite wounds: 875 mg oral q12hr or 500 mg oral q8hr for 3-5 days.
- *Erysipelas:* 875 mg oral q12hr or 500 mg oral q8hr for 7-10 days.
- *Diabetic foot:* mild to moderate, localized cellulitis: 2000 mg (2 extended release tablets) PO q12hr for 7-14 days.
- *Group A Streptococci infection:* 40 mg/kg/day PO divided q8hr for 10 days; not to exceed 2000 mg/day.
- *Tonsillitis and pharyngitis:* 500-875 mg oral q12h for 10 days.

Adverse effects: Diarrhea, mycosis, nausea, rash, vomiting, loose stool, candidiasis, vaginitis

Drug and food interactions: Amoxicillin decreases effects of BCG vaccine. Doxycycline, minocycline decreases effects of amoxicillin. Allopurinol may increase potential for hypersensitivity reactions to amoxicillin.

AMPICILLIN

Dosage form and strengths: *Capsules*: 250 mg, 500 mg. *Oral suspension*: 125 mg/5 ml, 250 mg/5 ml; *Injection*: 100 mg, 250 mg, 500 mg.

Indications: Urinary tract infections by *Escherichia coli* and *Proteus mirabilis,* exacerbation of chronic bronchitis and otitis media due to *Streptococcus pneumoniae* and *Haemophilus influenzae* and invasive salmonellosis.

Contraindications/Precautions: Cross allergy with cephalosporins, carbapenems. Adjust dose in renal failure. Pregnancy (B). Lactation: excreted in breast milk, use caution. Prolonged use associated with fungal or bacterial superinfection. Take on an empty stomach (1 hour before or 2 hours after a meal) with a full glass of water.

Dosage schedule: Oral: 0.25-1 g every 6 hours, at least 30 minutes before food.

- *Urinary tract infections:* 500 mg every 8 hours. Intramuscular injection or intravenous injections or infusion: 500 mg every 4-6 hours; higher doses in

meningitis. Child under 10 years: any route, half of adult dose.
Adverse effects: Erythema multiforme, exfoliative dermatitis, rash, urticaria, fever, seizure, diarrhoea, enterocolitis, glossitis, nausea, oral candidiasis, pseudomembranous colitis, stomatitis, vomiting, agranulocytosis, anaemia, eosinophilia, leukopenia, increased AST.
Drug and food interactions: BCG vaccine live, cholera vaccine, demeclocycline, doxycycline, lymecycline, minocycline, mycophenolate, oxytetracycline, tetracycline, typhoid vaccine live.

AMPICILLIN with SULBACTAM

Dosage form and strengths: *Powder for solution:* 1.5 g (ampicillin 1 g/ sulbactam 0.5 g), 3 g (ampicillin 2 g/sulbactam 1 g), 15 g (ampicillin 10 g/ sulbactam 5 g).
Indications: Gynecological infections, intra-abdominal infections, skin and skin structure infections, orbital cellulitis, pelvic inflammatory disease, pneumonia
Contraindications/Precautions: Hypersensitivity, patient with previous history of cholestatic jaundice/hepatic dysfunction associated with ampicillin sulbactam.Pregnancy (B). Lactation: excreted in breast milk. Caution in patients allergic to Cephalosporins and Carbapenems. Prolonged use is associated with fungal or bacterial superinfection. Hepatotoxicity reported; monitor hepatic function at regular intervals in patients with hepatic impairment. May cause severe skin reactions, such as toxic epidermal necrolysis (TEN), Stevens-Johnson syndrome, exfoliative dermatitis.
Dosage schedule:
- *Gynecologic infections:* 1.5 g (1 g ampicillin + 0.5 g sulbactam) to 3 g (2 g ampiccilin + 1 g sulbactam) IV/IM q6hr; not to exceed 12 g/day.
- *Intra-abdominal infections:* 1.5 g (1 g amoxicillin +0.5 g sulbactam) to 3 g (2 g ampicillin + 1 g sulbactam) IV/IM q6hr; not to exceed 12 g/day.
- *Skin and skin structure infections:* 1.5 g (1 g ampicillin +0.5 g sulbactam) to 3 g (2 g Ampicillin +1 g sulbactam) IV/IM q6hr; not to exceed 12 g/day.
- *Orbital cellulitis:* 3 g (2 g ampicillin + 1 g sulbactam) IV q6hr.
- *Pelvic inflammatory disease:* 3 g (2 g ampicillin + 1 g Sulbactam) IV q6hr.
- *Pneumonia (aspiration or community acquired):* 1.5 g (1 g ampicillin + 0.5 g sulbactam) to 3 g (2 g ampicillin +1 g sulbactam) IV q6hr for 5 or more days.

Adverse effects: Diarrhea, thrombophlebitis, rash, abdominal distension.
Drug and food interactions: Serious, use alternative: BCG vaccine, cholera vaccine. Ampicillin decreases effects of BCG, cholera vaccine by pharmacodynamic antagonism.

BACAMPICILLIN

Dosage form and strengths: *Tablets:* 400 mg
Indications: See under ampicillin.
Contraindications/Precautions: See under ampicillin.
Dosage schedule: 400-800 mg daily; child over 5 years: 200 mg 3 times daily. Uncomplicated gonorrhoea: single dose of 1.6 g together with probenecid 1 g.
Adverse effects: See under ampicillin.

BENZYL PENICILLIN (PENICILLIN G)

Dosage form and strengths: *Injection:* each vial of 150 mg, 300 mg and 600 mg. The powder should be dispersed in water for injection.

Indications: otitis media, meningitis, cellulitis, endocarditis, pneumococcal pneumonia, streptococcal infections, diphtheria.

Contraindications/Precautions: Hypersensitivity: serious and occasionally fatal reactions reported.Only deep IM administration; do not administer IV, SC, or IT. Do not inject near nerve or artery. Pregnancy (B). Caution during lactation and in neonates.

Dosage schedule: IM: 2.4-4.8 g daily in 4 divided doses; premature infant and neonate: 50 mg/kg daily in 2 divided doses; infant (1-4 weeks): 75 mg/kg daily in 3 divided doses; child (1 month-12 years): 100 mg/kg daily in 4 divided doses.

- *Bacterial endocarditis:* 7.2 g daily in 4-6 divided doses.
- *Meningitis:* 2.4 g every 4 hours; premature infant and neonate: 100 mg/kg daily in 2 divided doses.
- *Diphtheria:* 1200 mg IV q6hrly for 2weeks (along with diptheria antitoxin up to 1, 00,000 U IV)

Adverse effects: Skin rashes including maculopapular eruptions and exfoliative dermatitis, urticaria, serum-sickness like reactions, Jarisch-Herxheimer reaction, and pseudomembranous colitis.

CARBENICILLIN

Dosage form and strengths: *Injection*: 1 g, 5 g.

Indications: Infections due to *Pseudomonas aeruginosa* and *Proteus species*.

Contraindications/Precautions: Carbenicillin injection should be used immediately after preparation.

Dosage schedule: By slow intravenous injection or rapid infusion: 5 g every 4-6 hours; child: 250-400 mg/kg daily in divided doses. By intramuscular injection: 2 g every 6 hours; child: 50-100 mg/kg daily in divided doses.

Adverse effects: Hypokalaemia, prolonged bleeding time, prolonged prothrombin time, abnormal platelet aggregation and bleeding from gastrointestinal tract.

CLOXACILLIN

Dosage form and strengths: *Capsules:* 250 mg and 500 mg; *Injection:* 500 mg cloxacillin sodium in water for injection; *oral solution:* 125 mg/5 ml.

Indications: Exclusively for infection caused by *Staphylococcus aureus*.

Contraindications/Precautions: Hypersensitivity to penicillin, cephalosporins, imipenem. Concomitant live bacterial vaccines during initial treatment of severe infections. Neonates, pregnancy category B, lactation. Monitor PT if patient is taking warfarin concurrently; prolonged use may result in superinfection; injections should be used immediately after preparation.

Dosage schedule: Per oral: adult 500 mg 4 times daily, doubled in severe infection; by intramuscular injection: 250 mg every 4-6 hours, doubled in severe infection; by slow IV injection or infusion: 1-2 g every 6 hours; child up to 2 years: quarter adult dose; child 2-10 years: half adult dose.

Adverse effects: Abdominal pain, diarrhoea, nausea, anaemia, impaired LFTs, hypersensitivity, nephritis.

FLUCLOXACILLIN

Dosage form and strengths: *Capsule:* 250 mg, 500 mg.
Indications: Otitis externa, furunculosis, pneumonia, impetigo, osteomyelitis, staphylococcal endocarditis.
Contraindications/Precautions: See under benzylpenicillin.
Dosage schedule: Adult, oral: 250-500 mg every 6 hours, at least 30 minutes before food; child under 2 years: quarter adult dose; 2-10 year: half adult dose. Intramuscular injection: 250-500 mg every 6 hours; child under 2 years: quarter adult dose; 2-10 years: half adult dose.

- *Furunculosis:* 1 g every 6hour for 5 days
- *Endocarditis (in combination with another antibacterial) under 85 kg:* 8 g daily in 4 divided doses.

Adverse effects: See under benzylpenicillin.

PHENOXY METHYL PENICILLIN (PENICILLIN V)

Dosage form and strengths: *Oral solution*: 125 mg/5 ml, 250 mg/5 ml, *Tablet:* 250 mg, 500 mg
Indications: *Streptococcal* pharyngitis, actinomycosis, erysipelas, periodontal infections, recurrent rheumatic fever.
Contraindications/Precautions: Allergy to penicillins, cephalosporins, or imipenem. Pregnancy (B), lactation: excreted in breast milk; compatible with breastfeeding. IgE mediated anaphylactic reactions (e.g. urticaria, anaphylaxis) reported in patients with severe renal impairment and in patients with history of severe asthma. Use with caution in neonates; some dosage forms may contain sodium benzoate/benzoic acid, which may cause fatal toxicity (gasping syndrome) when used in large amounts. Monitor renal and hematologic systems. Prolong use may result in superinfection.
Dosage schedule:

- *Streptococcal pharyngitis:* 500 mg oral q12hr or 250 mg oral q6hr for 10 days.
- *Actinomycosis, mild:* 2-4 g/day oral divided q6hr for 8 weeks; surgical: 2-4 g/day oral divided q6hr for 6-12 months.
- *Erysipelas:* 500 mg oral q6hr.
- *Periodontal infections:* 250-500 mg oral q6hr for 5-7 days.
- *Recurrent rheumatic fever prophylaxis:* 250 mg oral q12hr.

Adverse effects: Diarrhea, nausea, oral candidiasis, vomiting, seizure, anemia, interstitial nephritis, hypersensitivity, anaphylaxis, positive coombs reaction.
Drug and food interactions: Penicillin V decreases effects of live BCG Vaccine (use alternative). Demeclocycline, doxycycline, minocycline decreases effects of penicillin V (use alternative).

PIPERACILLIN

Dosage form and strengths: *Powder for injection*: 2 g, 3 g, 4 g, 40 g.
Indications: Urinary tract infection (uncomplicated), community acquired pneumonia, acute cholangitis, uncomplicated gonorrhea, *Pseudomonas*

infections

Contraindications/Precautions: Allergy to penicillins, cephalosporins, imipenem.Pregnancy (B), lactation. Risk of bleeding complications, especially in renal impairment. Monitor renal, hepatic and especially hematopoietic functions during prolonged treatment. IV/IM preparation: reconstitute each gram w/5 ml NS, D5W or other compatible diluents. Slight darkening does not indicate potency loss. IV/IM administration: Slow direct injection over 3-5 min OR intermittent infusion in at least 50ml over 20-30min. IM at upper outer quadrant of buttock. Combined use of piperacillin/tazobactam and vancomycin may be associated with an increased incidence of acute kidney injury. Headache, fever, injection site pain may occur.

Dosage schedule:

- *Usual dosage range:* IV 3-4 g/dose q4-6hr; not to exceed 24 g/24hr IM 2-3 g/dose q6-12hr; not to exceed 24 g/24hr.
- *Urinary tract infection:* 6-8 g/day IV/IM (100 to 125 mg/kg/day) divided q6-12hr.
- *Community acquired pneumonia:* 6-8 g/day IV/IM (100-125 mg/kg/day) divided q6-12hr.
- *Acute cholangitis:* 4 g IVq6hr. Uncomplicated gonorrhea: 2 g once with 1 g probenecid 30 min before injection.
- *Pseudomonas infections:* 4 g IV/IM q4hr. In case of renal impairment: CrCL 20-40ml/min: 3-4 g q8hr. CrCl<20ml/min: 3-4 g q12hr.

Adverse effects: Seizure, rash, hemolytic anemia, interstitial nephritis, injection site pain, headache, fever, prolonged prothrombin time.

Drug and food interactions: Piperacillin increases effects and toxicity of antithrombin alfa, argatroban, bivaluridin, dalteparin, heparin, vancomycin. Azithromycin, doxycycline decreases effects of piperacillin. Piperacillin decreases effects of BCG vaccine.

PIPERACILLIN with TAZOBACTAM

Dosage form and strengths: *Powder for injection*: (2 g/250 mg)/vial: 2.25 g, (3 g/375 mg) *vial:* 3.375 g, (4 g/500 mg) *vial:* 4.5 g.

Indications: severe infections, nosocomial pneumonia, community acquired pneumonia, diverticulitis/intra-abdominal abscess/peritonitis complicated intra-abdominal infection, skin and soft tissue infection, malignant otitis externa

Contraindications/Precautions: Allergy to penicillin, cephalosporins, imipenem, beta lactamase inhibitors. Risk of bleeding complications, especially in renal impairment; discontinue if thrombocytopenia; discontinue if thrombocytopenia or bleeding occurs. Serious skin reactions reported, including Stevens-Johnson syndrome and toxic epidermal necrolysis, generalized exanthematous pustulosis; discontinue if reaction occurs.

Dosage schedule:

- *Severe infections:* intravenous: 3.375 g every 6 hr; total of 13.5 g (piperacillin [12 g] per tazobactam [1.5 g]) for 7-10 days; administer over 30 min.
- *Nosocomial pneumonia:* 4.5 g intravenous; add aminoglycoside; total of 18 g (piperacillin 16 g + tazobactam 2 g) for 7-14 days; continue aminoglycoside in *P. aeruginosa* patients.
- *Community acquired pneumonia:* intravenous: 3.375 g every 6 hr for 7-10

days.
- *Diverticulitis/intra-abdominal abscess/peritonitis:* intravenous: 3.375 g every 6 hr for 7-14 days or until clear.
- *Complicated intra-abdominal infection:* intravenous: 3.375 g every 6 hr for 4-7 days.
- *Skin and soft tissue infection:* intravenous: 3.375 g every 6-8 hr for 7-14 days.
- *Malignant otitis externa:* intravenous: 4.5 g every 8 hr then step down to oral ciprofloxacin 500-750 mg every 12 hr for total of 4-6 weeks.

Adverse effects: diarrhea, constipation, headache, insomnia, nausea, fever, oral candidiasis, rash, pruritus.
Drug and food interactions: see under piperacillin.

10.2.2 Protein synthesis inhibitors

10.2.2.1 Aminoglycosides

AMIKACIN
Dosage form and strengths: *Injection*: 50 mg/ml, 125 mg/ml and 250 mg/ml.
Indication: Serious gram-negative infections resistant to gentamicin.
Contraindications/Precautions: Pregnancy (D), hypersensitivity to any aminoglycoside, also see under gentamicin.
Dosage schedule: Adult: Once daily dosing IV infusion initially 15 mg/kg/day (max 1.5 g per dose / day), dose to be adjusted according to serum amikacin concentration: maximum 15 g per course. Child: Inj. 15 mg/kg daily in 2-3 doses, maximum of 1.5 g/day; Neonate: loading dose of 10 mg/kg then 15 mg/kg daily in 2 divided doses.
Adverse effects: See under gentamicin
Drug and food interactions: See under gentamicin.

GENTAMICIN
Dosage and strength: *Injection*: 10 mg/ml and 40 mg/ml; *Ear drops*: (0.3%) 3 mg/ml.
Indications: Urinary tract infections due to Pseudomonas, meningitis and other CNS infections, septicaemia and neonatal sepsis, endocarditis (with other antibiotics). Biliary tract disease, adjunct to listerial meningitis, prostatitis, surgical prophylaxis (including joint replacement surgery), medical labyrinthectomy, otitis externa.
Contraindications/Precautions: Hypersensitivity to any aminoglycosides, pregnancy (D). Breast-feeding, neonates, geriatric patients, mild renal disease, hearing deficits, myasthenia gravis, parkinson's disease, infant botulism, pseudomembranous colitis. Monitoring of blood level of gentamicin is advisable when used in high doses particularly in neonates, elderly and in those with renal impairment. To report if loss of hearing, ringing in the ears, feeling of fullness in the head.
Dosage schedule: By intramuscular or by slow intravenous injection over at least 3 minutes or IV infusion: 3-5 mg/kg daily (in divided doses every 8 hours). Child up to 2 weeks: 3 mg/kg every 12 hours; 2 weeks-12 years: 2 mg/kg every 8 hours. By intrathecal injection: 1 mg daily (increased if necessary

10.2.2.3 Lincosamide derivatives

CLINDAMYCIN

Dosage form and strengths: *Capsule:* 75 mg, 150 mg, 300 mg. *Topical lotion:* 10 mg/ 1 ml; *Gel:* 2%

Indication: Staphylococcal bone and joint infections such as osteomyelitis, Peritonitis, Intra-abdominal sepsis, Methicillin-resistant Staphylococcus aureus (MRSA), in bronchiectasis, bone and joint infections, and skin and soft-tissue infections, erysipelas or cellulitis in penicillin-allergic patients, treatment of mild to moderate pneumocystis pneumonia (in combination with primaquine), treatment of falciparum malaria (to be given with or following quinine), acne vulgaris, tonsillitis and pharyngitis (with penicillin allergy)

Contraindications/Precautions: Clostridium difficile associated diarrhea has been reported (mild to fatal colitis), hypersensitivity. Endocarditis prophylaxis: use only for high risk patients. Severe skin reactions including toxic epidermal necrolysis, drug reaction with eosinophilia and systemic symptoms (DRESS) and Steven Johnson Syndrome has been reported.

Dosage schedule:

- *Staphylococcal bone and joint infections such as osteomyelitis, Peritonitis, Intra-abdominal sepsis, Methicillin-resistant Staphylococcus aureus (MRSA), bronchiectasis, bone and joint infections, and skin and soft-tissue infections, erysipelas or cellulitis in penicillin-allergic patients (alternative to macrolides):* oral, Child: 3–6 mg/kg 4 times a day (max. dose 450 mg), Adult: 150–300 mg every 6 hours; increased if necessary up to 450 g every 6 hours if required, increased dose used in severe infection. Deep intramuscular injection or by intravenous infusion, adult: 0.6–2.7 g daily in 2–4 divided doses; increased if necessary up to 4.8 g daily, increased dose used in life threatening infection, single doses above 600 mg to be administered by intravenous infusion only, single doses by intravenous infusion not to exceed 1.2 g.
- *Pneumocystis pneumonia (mild to moderate form) (in combination with primaquine):* oral, adult: 600 mg every 8 hours.
- Falciparum malaria (to be given with or following quinine): oral, Child: 7–13 mg/kg every 8 hours (max. per dose450 mg) for 7 days. Adult: 450 mg every 8 hours for 7 days.
- *Acne vulgaris:* Adult: Apply 2% gel twice daily, to be applied thinly.
- *Tonsillitis and pharyngitis (with penicillin allergy):* 7 mg/kg/day in 3 divided doses (maximum 1.8 g per day) for 10days

Adverse effects: Abdominal discomfort, anaphylactoid reactions, antibiotic-associated colitis, diarrhoea (discontinue treatment), eosinophilia, exfoliative dermatitis, jaundice, leucopenia, nausea, oesophageal ulcers, oesophagitis, polyarthritis, pruritus,rash, Stevens-Johnson syndrome, taste disturbances, thrombocytopenia, toxic epidermalnecrolysis, urticarial, vesiculobullous dermatitis, vomiting, antibiotic-associated colitis.

Drug and Food Interaction: Clindamyin increases effects of atracurium by pharmacodynamic synergism. Risk of respiratory depression.

10.2.2.4 Oxazolidinone derivative

LINEZOLID

Dosage form and strengths: *Tablet:* 300 mg; *Oral suspension*: 100 mg/5 ml.
Indications: Pneumonia (when other antibacterials e.g. glycopeptide such as vancomycin cannot be used, initiated under specialist supervision). Complicated skin and soft tissue infection caused by gram positive bacteria when other antibacterials cannot be used (initiated under specialist supervision).
Contraindications/Precautions: Hypersensitivity, lactation, geriatric patients. Acute confusional states, bipolar depression, carcinoid tumors, history of seizures, pheochromocytoma, schizophrenia, thyrotoxicosis, uncontrolled hypertension. Visual function should be monitored regularly if treatment is required for longer than 28 days. Full blood counts are monitored weekly in patients who receive treatment for more than 10-14 days; have preexisting myelosuppression, have severe renal impairment, are receiving drugs that may have adverse effects on hemoglobin, blood counts, or platelet functions. Avoid consuming large amounts of tyramine rich foods like mature cheese, yeasts extracts, undistilled alcoholic beverages and fermented soya bean products. Plasma concentration of linezolid is reduced by rifampin leading to possible therapeutic failure of linezolid.
Dosage schedule: Oral, for adults: 600 mg every 12 hours usually for 10-14 days (maximum duration of treatment 28 days); Child: 10 mg/kg 8 hourly for 10-14 days; preterm infant: 10 mg/kg 12 hourly; by intravenous infusion 600 mg every 12 hours.
Adverse effects: Diarrhea, pruritus, pseudomembranous colitis, vaginal and oral candidiasis, taste disturbance, visual impairment, including blurred vision, visual field defect, color vision, change in visual acuity, pancytopenia
Drug and food interaction: Plasma concentration of linezolid is reduced by rifampicin, linezolid shouldn't be given with another MAOI, SSRIs, TCAs, or within 2 weeks of stopping another MAOI

10.2.2.5 Macrolides

AZITHROMYCIN

Dosage form and strengths: *Tablets:* 250 mg, 500 mg; *Dispersible tablets:* 100 mg; *Syrup:* 100 mg/5 ml, 200 mg/5 ml
Indications: Community-acquired pneumonia, lyme disease, respiratory-tract infections, otitis media, skin and soft tissue infections, non-gonococcal urethritis, multi-drug resistant typhoid tonsillitis and pharyngitis (with penicillin allergy).
Contraindications/Precaution: Contraindicated in impairment of hepatic function; Pregnancy category (B), breast-feeding, geriatric patients, renal/hepatic/cardiac disease; <6 months for otitis media; <2 years for pharyngitis, tonsillitis; QT prolongations, ulcerative colitis, pseudomembranous colitis, sunlight exposure. Use oral suspension 1 hour before or 2 hour after meal.
Dosage schedule:
• *For susceptible infections:* 500 mg once daily for 3 days; Child over 6

months, 10 mg/kg once daily for 3 days.
- *Endocarditis prophylaxis:* 15 mg/kg stat before surgical procedure.
- *Non-gonococcal urethritis:* 1 g as a single dose.
- *Typhoid:* 500 mg once daily for 7 days.
- *Lyme disease:* 500 mg daily for 7-10 days.
- *Acute otitis media:* 500 mg once then 250 mg once daily for 4 days.
- *Tonsillitis and pharyngitis (with penicillin allergy),* 500 mg oral daily for 5days

Adverse effects: Headache, drowsiness, constipation, arthralgia, disturbance in taste and vision, pancreatitis.

CLARITHROMYCIN

Dosage form and strengths: *Tablet:* 250 mg, 500 mg

Indications: Respiratory tract infections, mild to moderate skin and soft tissue infections, otitis media, *H. pylori* eradication in combination with proton pump inhibitor and metronidazole and in combination with proton pump inhibitor and amoxicillin, tonsillitis and pharyngitis

Contraindications/Precaution: Hypersensitivity to any macrolide, torsade de pointes, QT prolongation. Pregnancy (C), breast-feeding, geriatric patients, renal/hepatic disease. The drug should be used in pregnancy and breast-feeding if potential benefit outweighs risk.

Dosage schedule:
- *Respiratory tract infections:* Adult, oral: 250 mg every 12 hours for 7 days (severe infection, 500 mg every 12 hours for up to 14 days); Child: 15 mg/kg twice daily for 10 days.
- *H. pylori eradication in combination with a proton pump inhibitor and amoxicillin:* 500 mg twice daily.
- *H. pylori eradication in combination with a proton pump inhibitor and metronidazole:* 250 mg twice daily.
- *Tonsillitis and pharyngitis (with penicillin allergy):* oral: 250 mg every 12 hrs for 10 days.

Adverse effects: See under erythromycin; also tooth and tongue discoloration, headache, smell and taste disturbances, hyperhydrosis, insomnia.

ERYTHROMYCIN

Dosage form and strengths: *Tablets:* 100 mg, 250 mg, 500 mg; *Drops:* 100 mg/ml; *Oral Suspension:* 100 mg/5 ml

Indication: Lyme disease, chronic prostatitis, pertussis, diphtheria, prevention of recurrence of rheumatic fever, tonsillitis and pharyngitis (with penicillin allergy), acute laryngitis (bacterial).

Contraindications/Precautions: Hypersensitivity to any macrolide, history of cholestatic jaundice. Pregnancy (B), breast-feeding, hepatic disease, GI disease, QT prolongation, seizure disorder, myasthenia gravis. Hepatic function should be closely monitored in patients with a previous history of liver disease.

Dosage schedule:
- *Lyme disease:* 500 mg 4 times a day for 14-21 days.
- *Chronic prostatitis:* 250-500 mg 4 times a day, total daily dose may

alternatively be given in two divided doses. Increased to 4 g daily in divided doses in severe infections. Injection: 6.25 mg/kg every 6 hours for mild infections when oral treatment is not possible. Increased to 12.5 mg/kg every 6 hours in severe infections.

- *Prevention and treatment of diphtheria in non-immune patient:* 500 mg every 6 hours for 7 days, treat for further 10 days if nasopharyngeal swabs positive after 7 days treatment
- *Acne:* 500 mg twice daily for 3 months reduced to 250 mg twice daily for a further 3 months.
- *Early syphilis:* 500 mg 4 times daily for 14 days.
- *Uncomplicated non-gonococcal urethritis:* 500 mg twice daily for 14 days. Child: 20-50 mg/kg/day in divided dose q6hourly. Neonates: 20 mg/kg/day in divided dose q12 hourly.
- *Intestinal amoebiosis:* 30-50 mg/kg/day 4 times a dayfor 10-14 days.
- *Tonsillitis and pharyngitis (with penicillin allergy):* adult, oral: 500 mg 4 times a day for 10 days

Adverse effects: Nausea, vomiting, diarrhoea, skin rashes and fever. Cholestatic hepatitis is caused primarily by erythromycin estolate and rarely by erythromycin stearate or ethylsuccinate.

Drug and food interactions: Erythromycin potentiates the effects of carbamazepine, corticosteroids, digoxin, theophylline, tacrolimus and sildenafil.

10.2.2.6 Quinolones and fluoroquinolones

CIPROFLOXACIN

Dosage form and strengths: *Tablet:* 250 mg, 500 mg, 750 mg; *Injection:* 200 mg/100 ml. *Solution:* 0.2%

Indications: Uncomplicated and complicated UTIs, acute and chronic prostatitis, infective chronic airway disease, typhoid fever and gonorrhea, diffuse otitis externa

Contraindications/Precautions: Lactating mother, hypersensitivity to quinolones, in children. Pregnancy(C), children under 12 years. The drug should be used with caution in patients with epilepsy or history of epilepsy, hepatic or renal impairment, pregnancy and breast-feeding.

Dosage schedule:

- *Oral for respiratory tract infections:* 250-500 mg twice daily;
- *Urinary-tract infections:* 250-500 mg twice daily (100 mg twice daily for 3 days in acute uncomplicated cystitis in women);
- *Gonorrhea:* 500 mg as a single dose;
- *Chronic prostatitis:* 500 mg twice daily for 28 days;
- *Diffuse otitis externa:* oral 500 mg twice a day for 7 days.
- *Most other infections:* 500-750 mg twice daily;
- *Prophylaxis of meningococcal meningitis:* 500 mg as a single dose.
- *Surgical prophylaxis:* 750 mg 60-90 minutes before procedure; By intravenous infusion (over 30-60 minutes): 200-400 mg twice daily;
- *Urinary-tract infections:* 100 mg twice daily,gonorrhea: 100 mg as a single dose; Child: not recommended; but when benefit outweighs risk, oral: 10-30

mg/kg daily in 2 divided doses or by intravenous infusion: 8-16 mg/kg daily in 2 divided doses.

- *Ear infection:* 2-3 drops, twice daily for 7 days

Adverse effects: Nausea, vomiting, pancreatitis, tachycardia, hypotension, tinnitus, sweating, joint and cartilage damage, tendinitis and tendon rupture.
Drug and food interaction: Antacids reduce absorption of ciprofloxacin, concomitant use with warfarin prolongs PT.

LEVOFLOXACIN

Dosage form and strengths: *Tablets:* 250 mg, 500 mg. *Infusion:* 500 mg/ 100 ml.
Indications: Chronic prostatitis, urinary-tract infections, exacerbation of chronic bronchitis, community acquired pneumonia, skin and soft tissue infections, acute bacterial sinusitis, tonsillitis and pharyngitis (with penicillin allergy).
Contraindications/Precautions: Hypersensitivity to quinolones, children less than 18 years. To be used in caution in patients with convulsions and weakness of tendons, hypokalemia, hypomagnesemia, and renal dysfunction, pregnancy, breastfeeding, photosensitivity, acute MI, atrial fibrillation, colitis, dehydration. Monitor blood tests, electrolyte levels, sputum test, urine test and WBC count during treatment periodically. Assess for sign and symptoms of infection allergy and for bowel pattern. Administer 4 hours before or 2 hours after antacids, iron, zinc.
Dosage schedule:

- *Urinary-tract infections:* oral, adult: 250 mg daily for 7-10 days for complicated urinary tract infection. Increased in severe cases, (for 3 days in uncomplicated cases).
- *Exacerbation of chronic bronchitis:* oral, adult: 250-500 mg daily for 7-10 days.
- *Chronic prostatitis:* oral, adult: 500 mg once daily for 28 days.
- *Community-acquired pneumonia:* oral, adult: 500 mg once or twice daily for 7-14 days.
- *Skin and soft tissue infections:* oral, adult: 250 mg daily or 500 mg once or twice daily for 7-14 days.
- *Community acquired pneumonia:* oral, adult: 500 mg once or twice daily by intravenous infusion (over at least 60 minutes for each 500 mg). Child: 10-15 mg/kg single dose (Oral or IV). Tonsillitis and pharyngitis (with penicillin allergy): adult, oral: 500 mg oral once daily for 7days.

Adverse effects: See under ciprofloxacin. It also causes GI disturbance, insomnia, tachycardia, hypotension, hypoglycemia, and pneumonitis.

MOXIFLOXACIN

Dosage form and strengths: *Tablet:* 400 mg
Indication: Sinusitis, community acquired pneumonia, exacerbation of chronic bronchitis, mild to moderate pelvic inflammatory disease, tuberculosis.
Contraindication/Precautions: Hypersensitivity. Pregnancy(C), breastfeeding, hepatic/ cardiac/ renal impairment, epilepsy, uncontrolled

hypokalemia, prolonged QT interval, pseudomembranous colitis, diabetes mellitus. Assess ECG for cardiac status, assess BUN, creatinine, AST, ALT, electrolytes.

Dosage schedule:

- *Sinusitis:* oral dose for adults, 400 mg once daily for 7 days.
- *Community-acquired pneumonia:* oral dose for adults, 400 mg once daily for 7–14 days; by IV infusion for adults: 400 mg once daily for 7–14 days, to be given over 60 minutes.
- *Exacerbations of chronic bronchitis:* oral dose for adults, 400 mg once daily for 5–10 days.
- *Mild to moderate pelvic inflammatory disease:* oral dose for adults, 400 mg once daily for 14 days

Adverse effects: Angina, arrhythmias, constipation, flatulence, gastritis, hyperlipidemia, palpitations, sweating, vasodilation, photosensitivity, xerostomia, headache, dizziness

Food and drug interaction: Increase QT prolongation especially when it is combined with drugs that increase QT interval, probenecid increases serum moxifloxacin level, increase warfarin and cyclosporine effect, increase seizure risk when used with NSAIDs.

NALIDIXIC ACID

Dosage forms and strength: *Oral Suspension:* 300 mg/5 ml; *Tablets:* 500 mg

Indications: Urinary tract infections

Contraindications/Precautions See under Ciprofloxacin, children under 6 months.Renal and hepatic impairment.

Dosage schedule: 900 mg every 6 hours for 7 days, reduced in chronic infection to 600 mg every 6 hours; Child over 3 months maximum 50 mg/kg daily in divided doses; reduced in prolonged therapy to 30 mg/kg daily.

Adverse effects: Nausea, vomiting, abdominal pain, allergic reactions such as pruritus, urticaria, eosinophilia and fever, cranial nerve palsy, increase in intra-cranial pressure, metabolic acidosis, peripheral neuropathy, toxic psychosis

Drug and food interaction: Nitrofurantoin and nalidixic acid should not be used concurrently as antagonism occurs.

NORFLOXACIN

Dosage form and strengths: *Tablets:* 100 mg, 200 mg, 400 mg and 800 mg

Indications: Uncomplicated urinary tract infections, prophylaxis in recurrent urinary-tract infections, chronic prostatitis.

Contraindication/Precautions: Lactation, below 12 years, hypersensitivity to quinolones, tendon rupture. See under ciprofloxacin.

Dosage schedule:

- *Urinary-tract infections:* 400 mg twice daily for 7-10 days (for 3 days in uncomplicated lower urinary-tract infections).
- *Chronic relapsing urinary-tract infections:* 400 mg twice daily for up to 12 weeks; may be reduced to 400 mg once daily if adequate suppression within first 4 weeks.

- *Chronic prostatitis:* 400 mg twice daily for 28 days. Child: 10-15 mg/kg/day in two divided doses.

Adverse effects: Epiphora, see under ciprofloxacin.

Drug interactions: see under ciprofloxacin.

OFLOXACIN

Dosage form and strengths: *Tablets*: 100 mg and 200 mg, *I.V. infusion:* 200 mg/100 ml. *Solution:* 0.3% (3 mg/ml)

Indication: Uncomplicated urinary-tract infections, acute or chronic prostatitis, infective chronic airway disease, multidrug resistant TB, PID, septicemia, skin, and soft tissue infection, acute otitis externa, otitis media.

Contraindications/Indications: See under ciprofloxacin.

Dosage schedule:

- *Skin and soft tissue infections:* 400 mg twice a day for 10 days.
- *PID:* 400 mg twice a day for 14 days.
- *Septicemia:* Inj. adult 200 mg twice a day, increased to 400 mg twice a day for severe complicated infections. Each 200 mg given over at least 30 mins.
- *Urinary-tract infections:* 200-400 mg daily preferably in the morning, increased if necessary in upper urinary-tract infections to 400 mg twice daily.
- *Lower respiratory-tract infections:* 400 mg daily preferably in the morning, increased if necessary to 400 mg twice daily.
- *Uncomplicated gonorrhea:* 400 mg as a single dose.
- *Non-gonococcal urethritis:* 400 mg daily in single or divided doses for 7 days.
- *By intravenous infusion for complicated urinary-tract infection:* 200 mg daily (over at least 30 minutes for each 200 mg).
- *Lower respiratory-tract infections:* 200 mg twice daily. Child: Oral 15-20 mg/kg/day in two divided doses
- *Acute otitis externa:* 10 drops in affected ear once daily for 7 days;
- *Otitis media:* same as in acute otitis externa for 14 days

10.2.2.7 Sulphonamides

COTRIMOXAZOLE (SULPHAMETHOXAZOLE and TRIMETHOPRIM)

Dosage form and strengths: *Oral suspension*: 40 mg trimethoprim and 200 mg sulphamethoxazole per 5 ml. *Tablets:* 80 mg trimethoprim plus 400 mg sulphamethoxazole, 40 mg trimethoprim plus 200 mg sulphamethoxazole, 160 mg trimethoprim plus 800 mg sulphamethoxazole. *Dispersible tablets:* 20 mg trimethoprim plus 100 mg sulphamethoxazole.

Indication: Uncomplicated lower urinary tract infection, bacterial prostatitis, exacerbation of chronic bronchitis due to *H. influenzae* and *Strep. pneumoniae*, acute otitis media in children and acute maxillary sinusitis in adults due to *H. influenzae* and *Strep. pneumoniae*, *Pneumocystiscarinii* pneumonia.

Contraindication/Precaution: Blood donors, avoid in infants less than 6 weeks, G-6PD deficiency, hypersensitivity, pregnancy, lactation. Monitor for pseudomembranous colitis. Risk of crystalluria can be decreased by maintaining urinary output of at least 1.5 liters daily.

Adverse effects: Abdominal pain, diarrhea and rash. In heavily infected children, roundworms may be expelled through the mouth and nose, since mebendazole kills these worms slowly and cause them to migrate.
Drug and food interaction: Decrease mebendazole effect by carbamazepine, hydantoins. Increase absorption by high fat meal.

MILTEFOSINE

Dosage form and strengths: *Capsule:* 50 mg
Indications: Visceral and cutaneous leishmaniasis.
Contraindication/Precaution: Pregnancy (D), breast-feeding (5 months following therapy). Hepatic impairment. Advise females to use additional non-hormonal or alternative methods of effective contraception. Encourage fluid intake to avoid volume depletion.
Dosage schedule: Oral 100 mg daily (patients weighing more than 25 kg) for 28 days; child 2.5 mg/kg daily.
Adverse effects: Vomiting, diarrhoea, rise in hepatic transaminases and serum creatinine (reversible).
Drug and food interaction: Combination therapy with paromomycin or Amphotericin B prevents development of resistance to this drug compared to individual use.

NICLOSAMIDE

Dosage form and strengths: *Tablet:* (chewable) 500 mg.
Indications: Tapeworm infections (*Taenia solium, Taenia saginata, Hymenolepsis nana* and *Diphyllobothrium latum*).
Contraindications/ Precaution: Hypersensitivity, alcohol. Lactation, pregnancy (B), elderly, children. It is inactive against cysticercosis cellulose and danger of cysticersosis must be considered when used in *T. solium* infection. Check patient for the history of allergy before starting treatment. Avoid alcohol intake during treatment with niclosamide.
Dosage schedule:

- *Taenia solium:*- Adult and child over 6 years 2 g as a single dose after a light breakfast or meal followed by a purgative after 2 hours; child under 2 years 500 mg, 2-6 years 1 g.
- *T. saginata* and *Diphyllobothrium latum*: as for *T. solium* but half the dose may be taken after breakfast or meal and the remainder 1 hours later followed by a purgative 2 hours after last dose.
- *Hymenolepis nana*: adult and child over 6 years: 2 g as a single dose on first day then 1 g daily for 6 days; child under 2 years:500 mg on first day then 250 mg daily for 6 days, 2-6 years: 1 g on first day then 500 mg daily for 6 days.

Adverse effects: Nausea, vomiting, abdominal discomfort, anorexia, diarrhoea and pruritus.
Drug and food interaction: Niclosamide is soluble in alcohol which enhances its absorption raising the possibility of dose related adverse effects.

PENTAMIDINE

Dosage form and strengths: *Injection:* 200 mg and 300 mg vials.

Indications: Visceral leishmaniasis, *Pneumocystis* pneumonia.

Contraindications/Precautions: Hypersensitivity. Pregnancy (C), breastfeeding, children, blood dyscrasias, cardiac/renal/hepatic disease, diabetes mellitus, hypoglycemia, hypocalcemia, hypo/hypertension, anemia.

Dosage schedule:

- *Visceral leishmaniasis:* deep intramuscular injection 3-4 mg per kg of body weight on alternate days to a maximum of 10 injections; course may be repeated if necessary.
- *Pneumococcal pneumonia, Pneumocystis pneumonia*: intravenous infusion, 4 mg per kg of body weight daily for at least 14 days.

Adverse effects: Rash, abnormal liver function tests, hypotension, hyperglycaemia, hypoglycaemia, thrombocytopenia, acute renal failure, hyperkalaemia, megaloblastic anaemia, acute pancreatitis and pain at site of injection. The drug should be used cautiously in the presence of hypertension, hypotension, diabetes, kidney disease.

Drug and food interaction: Increases potential for nephrotoxicity:- aminoglycosides, amphotericin B, cisplatin, NSAIDs, vancomycin. Fetal dysrhythmias:- erythromycin IV. Increase QT prolongation- class III/IA antiarrhythmic, some phenothiazines, β-agonists, local anesthetics, tricyclic antidepressants, haloperidol, chloroquine, CYP3A4 inhibitors (amiodarone, clarithromycin, erythromycin), CYP3A4 substrates (methadone, quetiapine, quinidine, risperidone, ziprasidone). Increase myelosuppression:-anti-neoplastics, radiation.

PIPERAZINE

Dosage form and strengths: *Elixir:* 750 mg/5 ml. *Tablet:* 300 mg

Indication: Ascariasis, enterobiasis.

Contraindications/Precautions: Epilepsy, hepatic or renal impairment, in patients who are hypersensitive to this drug. Children, pregnancy (B), breast feeding, only used if alternative medications are not available. Stool examination may be required before treatment and approximately 2 weeks following treatment with piperazine to determine efficacy or proof of care. Perianal examination is done 1 week following treatment with piperazine especially in patients with persisting symptoms to determine efficacy or proof of cure. In some patients, pinworms may return after treatment with piperazine, washing all bedding and night clothes after treatment may help to prevent this; because of high probability of transfer of pinworms it is often recommended that all members of household be treated concurrently

Dosage schedule:

- *Ascaraisis:* up to 3.5 g per day: Child 75 mg/kg body weight per day for two consecutive days. Treatment may be repeated after one week for heavy infection.
- *Enterobiasis:* up to 2.5 g per day: Child 65 mg/kg body weight per day for seven consecutive days.

Adverse effects: Nausea, vomiting, mild diarrhoea, abdominal pain and urticaria.

Drug and food interaction: Piperazine may enhance the effects of phenothiazines and increase the risk of seizures. Pyrantel pamoate and

Heart diseases (withdrawn if conduction disturbances occur), mucocutaneous diseases, predisposition to QT interval prolongation. Monitor ECG before and during treatment. Preparation protect from light.

Dosage schedule: 20 mg/kg daily (maximum 850 mg) for at least 20 days by intramuscular or intravenous injection. Skin lesions are treated for 10 days.

Adverse effects: Headache, skin rashes, vomiting and abdominal pain. Rapid intravenous injection can result in severe cough, vomiting and even cardiovascular collapse.

10.5 Antileprotic drugs

CLOFAZIMINE

Dosage form and strengths: *Capsule*: 50 mg, 100 mg

Indication: Multibacillary leprosy in combination with rifampicin and dapsone (3 drug regimen), Type II lepra reaction.

Contraindications/Precautions: Hepatic and renal impairment. Pregnancy, old age, hepatic and renal impairment. Observe patient for history of allergy to clofazimine, GI problems and diarrhea. Observe patient physically (skin lesions, colour, turgor, texture), ocular examination, bowel sounds, stool color, and culture of the lesion. Antibacterials: increased risk of ventricular arrhythmias when given with bedaquiline. Avoid if persistent abdominal pain and diarrhea, may discolor soft contact lenses, skin discoloration of infants.

Dosage schedule: Oral, adult: 300 mg once a month (supervised administration) and 50 mg daily (self-administered) for 1 year.

Adverse effects: Acne like eruptions, dry eyes, dry skin, phototoxicity, nausea, vomiting, and headache, red discoloration of skin, faeces and urine.

DAPSONE

Dosage form and strengths: *Tablets:* 50 mg and 100 mg.

Indications: Leprosy, dermatitis herpetiformis, prophylaxis of *Pneumocystis* pneumonia, preventing recurrence of rhinosporidiosis.

Contraindications/Precaution: Cardiac disease, G6PD deficiency, pulmonary disease,

Dosage schedule:

- *Multi-bacillary leprosy:* rifampicin 600 mg and clofazimine 300 mg are both given once a month under supervision together with dapsone 100 mg and clofazimine 50 mg both daily in self-administered doses for 1 year. Doses of all 3 agents are reduced in children and in those aged 10-14 years and adult weighing less than 35 kg, daily dose of dapsone 50 mg or 1-2 mg/kg.
- *Paucibacillary leprosy:* rifampicin 600 mg under supervision once a month and dapsone 100 mg self-administered daily, both agents are given for 6 months. Doses are reduced in children and low body weight patient as for multibacillary leprosy.
- *Prophylaxis and treatment of Pneumocystis pneumonia:* adult 100 mg daily.

Adverse effects: Anorexia, nausea, vomiting, headache, pruritus and haemolytic anaemia (in G6PD deficiency).

Drug and food interaction: Pregnancy- Neonatal hemodialysis and methemoglobinemia reported in third trimester. Folic acid should be given to

hypoglycemia, nephrotoxicity.

Drug and food interaction: Additive vagolytic effect with anticholinergic. Increase cardiac depression with other anti-dysrhythmic, phenothiazines, reserpine. Increase effects of neuromuscular blockers, digoxin, warfarin, tricyclics, and propranolol. Increase QT prolongation by macrolides, quinolones, tricyclics, procainamide, antipsychotic

SULFADOXINE and PYRIMETHAMINE

Dosage form and strengths: *Tablet:* each containing 500 mg of sulfadoxine and 25 mg of pyrimethamine.

Indications: Combination with other antimalarials in malaria caused by P. Falciparum only.

Contraindications/Precautions: Acute porphyrias, sullfonamide allergy, pregnancy, breastfeeding. Asthma, avoid in blood disorders, avoid in infants under 6 weeks, elderly, G6PD deficiency, predisposition to folate deficiency, predisposition to hyperkalemia in adults. **:** Monitor blood counts on prolonged treatment.

Dosage schedule: *Treatment:* adult 3 tablets as single dose; child: 1.25 mg/kg body weight of pyrimethamine and 25 mg/kg body weight of sulphadoxine.

Adverse effects: Diarrhoea, headache, hyperkalemia, nausea and rash.

Drug and food interaction: Folic acid + (sulfadoxine and pyrimethamine): decrease effects of each other. Sulfadoxine and pyrimethamine increases levels of chlorpromazine by decreasing metabolism.

Patient information: To maintain adequate fluid intake to avert crystalluria or stone formation.

10.7 Anti-tubercular drugs

10.7.1 First line drugs

ETHAMBUTOL

Dosage form and strengths: *Tablet:* 100 mg, 400 mg, 600 mg, 800 mg

Indications: Tuberculosis in combination with other drugs.

Contraindication/Precautions: Children less than 5 years, hypersensitivity, optic neuritis, severe renal impairment. Pregnancy B, breast-feeding, renal disease, diabetic retinopathy, cataracts, ocular defects, hepatic and hematopoietic disorders, severe renal impairment. Renal function should be checked before treatment. Visual acuity should be tested by Snellen chart before treatment with ethambutol. In children routine ophthalmological monitoring is recommended.

Dosage schedule: 15-25 mg/kg daily.

Adverse effects: Optic neuritis (reversible) resulting in decrease in visual acuity and loss of ability to differentiate red from green. The effect is dose dependent and occurs rarely on dose of 15 mg/kg given daily. The other adverse effects are skin rash, drug fever, pruritus, hyperuricemia, joint pain and gastro-intestinal upset.

Drug and food interaction: Ethambutol decreases effects of live BCG vaccine by pharmacodynamic antagonism Aluminium hydroxide :increases levels of

etahmbutol by cation binding in GI tract. Ethambutol decreases effects of allopurinol- hyperuricemia reported with ethambutol and precipitation of gout reported. Neurotoxicity- increases by other neurotoxic drugs.
Patient information: To report any visual changes, rash, hot swollen and painful joints, numbness, tingling of extremities.

ISONIAZID

Dosage form and strengths: *Tablets:* 100 mg and 300 mg.
Indications: Tuberculosis in combination with other drugs, TB prophyllaxis.
Contraindications/Precautions: Hypersensitivity, acute hepatic disease, drug induced hepatic disease. Pregnancy (C), renal impairment, diabetic retinopathy, cataracts, ocular defects, IV drug users, > 35 years, postpartum, HIV, neuropathy, alcoholism, acute porphyria, excessive alcohol intake and history of liver disease should be assessed monthly for hepatic-damage. Pyridoxine (25-100 mg) per day should be given for the prevention of peripheral neuropathy. The drug should be used with caution in patients with impaired liver and kidney function. Hepatic and renal functions should be checked before treatment. Those with alcohol dependence should have frequent checks for hepatic function, particularly in the first two months.
Dosage schedule: 300 mg daily; Child: 10 mg/kg daily (maximum 300 mg daily)
Adverse effects: Peripheral neuritis is the most common adverse effect, higher when high doses are used; skin rash, ataxia, dizziness, optic neuritis and hepatic damage.
Drug and food interaction: Increased toxicity when given with tyramine containing foods, alcohol, cycloserine, ethionamide, rifampin, carbamazepine, phenytoin, benzodiazepines, meperidine. Increased risk of serotonin syndrome when given with SSRIs, SNRIs. Aluminium antacids decreases its absorption. Decreases effectiveness of BCG vaccine and ketoconazole.
Patient information**:** To avoid alcohol while taking the drug as it may increase the risk of hepatic disease.

PYRAZINAMIDE

Dosage form and strengths: *Table:t* 400 mg/500 mg/750 mg/1 g.
Indications: Tuberculosis
Contraindications/Precautions: Acute attack of gout, pregnancy (C), lactation. Diabetes, gout, old age. RFT and LFT needs to be assessed at onset of treatment. Monitor for presence of gout.
Dosage schedule: Adult <50 kg: 1.5 g/day, adult > 50 kg: 2 g/day, Child: 25-35 mg/kg daily.
Adverse effects: Hepatotoxicity is the serious adverse effect. Anorexia, nausea, vomiting and arthralgia are less common. Rise in serum transaminase concentrations are common during the early phase of treatment and return to normal despite continuation of treatment, in most of cases. Hyperuricemia, sideroblastic anaemia, thrombocytopenia.
Drug and food Interactions: Rifampin+Pyrazinamide: either increases toxicity of the other by pharmacodynamic synergism. Additive hepatotoxicity.

RIFAMPIN

Dosage form and strengths: *Capsules:* 150 mg, 300 mg and 450 mg. *Tablets:* 150 mg, 300 mg

Oral Suspension: 100 mg/ 5 ml,

Indications: Short course chemotherapy of tuberculosis and leprosy, prevention of meningococcal meningitis and *H. influenzae,* atrophic rhinitis.

Contraindication/Precautions: Hypersensitivity to this product and rifamycin, active *Nisseria meningitis* infection, acute porphyrias. Pregnancy (C), breastfeeding, children less than 5 years, hepatic disease, blood dyscrasias.

Dosage schedule:

- *Brucellosis, legionnaire's disease and serious staphylococcal infections:* in combination with other drugs, oral or by i.v infusion: 0.6-1.2 g daily in 2-4 divided doses.
- *Tuberculosis, in combination with other drugs:* adult (more than 50 kg) 600 mg daily for 6 months in category-I and for 8 months in category-II; child: 15 mg/kg daily for 6 months in category-I and for 8 months in category-II.
- *Leprosy, multibacillary leprosy:* (3 drugs regimen), 600 mg once monthly, supervised (450 mg for those weighing less than 35 kg.)
- *Paucibacillary leprosy:* (2 drugs regimen) 600 mg once monthly supervised (450 mg for those weighing less than 35 kg).
- *Prophylaxis of meningococcal meningitis:* 600 mg every 12 hours for 2 days; child 3 months - 1 year 5 mg/kg, over 1 year 10 mg/kg every 12 hours for 2 days.
- *Prophylaxis of H. influenza type B:* 600 mg once daily for 4 days; child 1-3 months 10 mg/kg once daily for 4 days, over 3 months 20 mg/kg once daily for 4 days (maximum 600 mg daily).
- *Atrophic rhinitis: oral:* 600 mg once daily for 12 weeks.

Adverse effects: Anorexia, nausea, vomiting, diarrhoea, "flu" like syndrome characterized by fever, malaise, headache, chills, skin rashes, transient rise in serum bilirubin and transaminases, respiratory symptoms, thrombocytopenic purpura and orange/red body secretions.

Adverse effect are more likely to occur during intermittent therapy, however the monthly schedules in leprosy appears to be devoid of this risk.

Drug and food interactions: Drug induces hepatic enzymes which accelerate the metabolism of several drugs including corticosteroids, digitalis glycosides, estrogens and oral hypoglycemic. Increase hepatotoxicity with isoniazid. Do not use with protease inhibitors.

Patient information: To avoid alcohol because hepatotoxicity may occur. Urine, saliva, faeces, sputum, sweat, tears may be colored red-orange. Soft contact lenses may be permanently stained

STREPTOMYCIN

Dosage form and strengths: Injection: 1 g and 0.75 g streptomycin sulphate.

Indications: Tuberculosis resistant to other treatment in combination with other drugs, brucellosis and bacterial endocarditis.

Contraindications/Precautions: Pregnancy and over 40 years. Should be avoided during pregnancy and in patient over 40 years whenever possible.

Drug and food interaction: See under gentamicin.
Dosage schedule: Adult: 0.75 g daily; child 15-20 mg/kg/day IM
Adverse effects: Vestibular damage is the main adverse effect, the risk increases with dose and age (above 40 years), skin rashes, drug fever, nephrotoxicity, hypersensitivity antibiotic associated colitis.

10.7.2 Second line drugs

BEDAQUILINE

Dosage form and strengths: *Tablet:* 100 mg
Indications: combination regimen to treat pulmonary MDR-TB infection when other effective treatment regimens are not available.
Contraindications/Precautions: Alcoholism, bradycardia, breastfeeding, arrhythmias, cardiac impairment, children, coronary artery disease, diabetes mellitus, females, geriatric, heart failure, hepatic impairment, hypertension, hypocalcemia, hypokalemia, hypomagnesemia, malnutrition, MI, pregnancy (B), syncope, thyroid disease. Obtain ECG before starting treatment and then at least monthly during treatment or more frequently if concomitant use with other drugs known to prolong QT interval.
Dosage schedule: *MDR-Pulmonary TB, in combination with other drugs.* Adult: Initially 400 mg once daily for 2 weeks, then 200 mg 3 times a week for 22 weeks, intervals of at least 48 hours between each dose, continue appropriate combination therapy after bedaquilne
Adverse effects: Arthralgia, diarrhea, dizziness, headache, myalgia, nausea, QT interval prolongation, vomiting.
Drug and food interactions: Increase QT prolongation: other drugs that prolong QT (class IA/III antidysrhythmics, some phenothiazines, beta agonists, local anesthetics, tricyclics, haloperidol, chloroquine, droperidol; CYP3A4 inhibitors (amiodarone, clarithromycin, erythromycin); CYP3A4 substrates (methadone, pimozide, quetiapine, quinidine, risperidone, ziprasidone)
Increase adverse reactions: lopinavir/ritonavir. Increase bedaquiline effect: strong CYP3A4 inhibitors, avoid use over 14 days. Decrease bedaquiline effect: strong CYP3A4 inducers, avoid concurrent use.

CAPREOMYCIN

Dosage form and strengths: Powder for injection, 1 g in vial.
Indications: TB resistant to 1st line drugs in combination with other drugs
Contraindications/Precautions: Hypersensitivity, anxiety, epilepsy, psychotic reactions, porphyria. Auditory impairment, hepatic and renal impairment, breast feeding, children, pregnancy, old age. Monitor renal, hepatic, auditory and vestibular function and electrolytes.
Dosage schedule: *TB resistant to first line drugs in combination with other drugs:* By deep IM injection Adult: 1 g daily (max. per dose 20 mg/kg) for 2-4 months, then reduced to 1 g 2-3 times a week.
Adverse effects: Induration at injection site, changes in liver function tests, electrolyte disturbances, hearing loss with tinnitus and vertigo, hypersensitivity reactions, leucocytosis, leucopenia, nephrotoxicity,

neuromuscular, block after large doses, pain at injection site, rashes, thrombocytopenia, urticaria.
Drug and food interactions: Antibacterials: increased risk of nephrotoxicity when given with colistin or polymyxins; increased risk of nephrotoxicity and ototoxicity when given with aminoglycosides or vancomycin.
Cytotoxics: increased risk of nephrotoxicity and ototoxicity when given with platinum compounds.

CYCLOSERINE

Dosage form and strengths: Capsule 250 mg
Indications: TB resistant to 1st line drugs in combination with other drugs.
Contraindications/Precautions: Alcohol dependence, depression, epilepsy, psychotic states, severe anxiety. Blood concentration monitoring required especially in renal impairment or if dose exceeds 500 mg daily or if signs of toxicity; Blood concentration should not exceed 30 mg/litre. Monitor hematological, renal and hepatic function.
Dosage schedule: *TB resistant to first line drugs in combination with other drugs:* Initially 250 mg every 12 hours for 2 weeks, then increased if necessary up to 500 mg every 12 hours, dose to be increased according to blood concentration and response. Child: 10 mg/kg/day in 2 divided doses.
Adverse effects: Allergic dermatitis, changes in liver function tests, confusion, convulsions, depression, dizziness, megaloblastic anemia, psychosis, rashes, tremor, vertigo.
Drug and food interactions: Alcohol: increased risk of convulsions when cycloserine given with alcohol. Increased risk of CNS toxicity when cycloserine given with isoniazid.

ETHIONAMIDE

Dosage form and strengths: *Tablet*: 125 mg, 250 mg
Indications: Second line drug for TB
Contraindications/Precautions: Hypersensitivity, hepatic impairment, porphyria. Pregnancy, lactation, psychiatric cases or patient with psychosis.
Dosage schedule: Initiate dose at 250 mg/day for 1-2 days then increase to 250 mg twice daily for 1-2 days with gradual increase to highest tolerated dose; 750 mg/day average dose; not to exceed 100 mg/day in 3-4 divided dose.
Adverse effects: Nausea, vomiting, diarrhea, anorexia, postural hypotension, depression, dizziness, drowsiness, headache, peripheral neuropathy, psychosis, photosensitivity, rash, excessive salivation, gynaecomastia, hypoglycemia, impotence, thrombocytopenia, hepatitis, optic neuritis, visual changes.
Drug and food interactions: Ethionamide and cycloserine increase the toxicity of each other. Increases level of isoniazid. Decreases the effect of magnesium oxide, sodium picosulfate.

KANAMYCIN

Dosage form and strengths: *Powder/ solution*: for injection 1 g in vial/ ampoule

15.1 Diagnostic agents

TUBERCULIN, PURIFIED PROTEIN DERIVATIVE (PPD)

Dosage form and strength: *Solution:* 2 units is equivalent to 0.1 mL of 20 units/mL strength, 10 units is equivalent to 0.1 mL of 100 units/mL strength.
Indications: Test for hypersensitivity to tuberculoprotein (Mantoux test)
Contraindications/Precautions: It should not be used within 4 weeks of receiving live viral vaccination, viral and bacterial infections (including HIV and severe tuberculosis), malignant disease, corticosteroid or other immunosuppressive therapy, malnourished patients, elder patients. Mantoux test response to tuberculin may be suppressed by live viral vaccines, viral infection, sarcoidosis, corticosteroid therapy, or immunosuppression due to disease or treatment. Let the prescriber or your physician know if any of the above symptoms do not go away or worsen.
Dosage schedule: Intradermal injection: 10 units for 1 dose.
Adverse effects: Headache, nausea, malaise, rash, itch, vesicle formation, ulceration, necrosis or pain at the administered site.
Drug and food interactions: Corticosteroids (e.g. prednisone), immunosuppressives (e.g. certain cancer medicines, cyclosporine), or live vaccines (e.g. measles) because the effectiveness of tuberculin purified protein derivative (PPD) may be decreased.

15.2 Sera and immunoglobulins

ANTI-D IMMUNOGLOBULIN (HUMAN)

Dosage form and strength: *Solution:* 300 µg/2 ml.
Indications: Prevention of formation of antibodies to rhesus positive blood cells in rhesus negative patients.
Contraindications/Precautions: Known hypersensitivity, cautiously used in rhesus positive patients for treatment of blood disorders, rhesus negative patients with anti D antibodies in the serum.
Use only those immunoglobulin preparations which have been screened adequately.
Dosage schedule:

- *Rhesus-negative woman for prevention of D sensitization following abortion or birth of rhesus-positive infant:* Deep intramuscular injection: 500 units immediately or within 72 hours; for transplacental bleed in excess of 4 ml fetal red cells, extra 100-125 units per ml fetal red cells.
- *Following any potentially sensitizing episode (e.g. still birth):* Deep intramuscular injection up to 20 weeks gestation 250 units per episode, after 20 weeks of gestation: 500 units. Immediately or within 72 hours of such event.
- *Following $Rh_o(D)$ incompatible blood transfusion:* Deep intramuscular injection: 100-125 units/ ml transfused rhesus-positive red cells.
- *To rhesus-negative woman for prevention of Rh (D) sensitization antenatal prophylaxis:* deep intramuscular injection, 500 units to be given at weeks 28 and 34 of pregnancy, if infant rhesus-positive, a further dose is still needed immediately or within 72 hours of delivery, subcutaneous route used for patients with bleeding disorders.

Adverse effects: Soreness at the site of injection, fever, anemia, intravascular

Contraindications/Precautions: Not recommended for infants under 9 months of age; pregnancy; individuals with severe immunodeficiency or severe egg allergy, acute illness

Dosage schedule:

- *Immunization of children against yellow fever:* deep subcutaneous/ intramuscular injection, infant (9–12 months): 0.5 ml as a single dose.
- *Immunization of travellers and other at-risk individuals against yellow fever:* deep subcutaneous/ intramuscular injection: adult and child (over 9 months): 0.5 ml as a single dose.

Adverse effects: Headache, myalgia, weakness; very rarely encephalitis (infants more susceptible), viscerotropic disease, multiple organ failure (the elderly more susceptible).

EPI Schedule of Nepal, 2017

S.N.	Age (from birth)	Immunization	Prophylaxis/Indication
1.	At birth	B.C.G.	Tuberculosis
2.	6 weeks	Rota (first dose)	Rota virus induced diarrhea
		Polio (first dose)	Polio
		FIPV (first dose)	Polio
		PCV (first dose)	Pneumococcal Pneumonia and other infection
		DPT-Hep B- Hib (first dose)	Diphtheria, pertussis, tetanus, hepatitis B and hemophilus influenza
3.	10 weeks	Rota (second dose)	Rota virus induced diarrhea
		Polio (second dose)	Polio
		PCV (second dose)	Pneumococcal Pneumonia and other infection
		DPT-Hep B- Hib (second dose)	Diphtheria, pertussis, tetanus, hepatitis B and hemophilus influenza
4.	14 weeks	Polio (third dose)	Polio
		FIPV (second dose)	Polio
		DPT-Hep B- Hib (third dose)	Diphtheria, pertussis, tetanus, hepatitis B and hemophilus influenza
5.	9 months	PCV (third dose)	Pneumococcal Pneumonia and other infection
		Measles and rubella (first dose)	Measles and rubella
6.	12 months	Japanese encephalitis	Japanese encephalitis
7.	15 months	Measles and rubella (second dose)	Measles and rubella

of artemether with lumefantrine advises avoid concomitant use)
- Chloroquine, Moxifloxaxin, Haloperidol, Droperidol: Increased risk of ventricular arrhythmias
- Digoxin*: Plasma concentration of digoxin increased
- Mefloquine*: Increased risk of convulsions, but should not prevent the use of intravenous quinine in severe cases
-

RABIES VACCINE
- Hydrocortisone, Methylprednisolone, Triamcinolone, Betamethasone, Dexamethasone: Associated with a diminished or suboptimal immunologic response due to antibody inhibition
- Chloroquine: Reduces the antibody response to primary immunization with intradermally administered rabies vaccine.

RANITIDINE
- Tolazoline: Antagonizes the effect of ranitidine
- Posaconazole, Azatanavir: Reduces the concentration
- Loperamide: enhance the gastrointestinal absorption or inhibit the metabolism of loperamide
- Theophylline, Glipizide: Increases the concentration
- Metformin: Decreases the excretion of metformin

RIFAMPICIN
- Antacids (Aluminium hydroxide; Magnesium hydroxide): Reduced absorption of rifampicin
- Contraceptives*, Oral, Propranolol, Prednisolone*, Phenytoin*, Nifedipine*, Methadone, Haloperidol*, Verapamil* Warfarin*, Medroxyprogesterone*, Quinidine*, Levothyroxine, Norethisterone*, Levonorgestrel*, Indinavir*, Hydrocortisone*, Fluconazole*, Diazepam, Dexamethasone*, Cyclosporin*, Chloramphenicol: Accelerated metabolism
- Saquinavir*, Nevirapine*, Nelfinavir*, Lopinavir*, Efavirenz, Dapsone: Reduced plasma-dapsone concentration: Reduced plasma concentration
- Zidovudine: Avoidance of rifampicin advised by manufacturer of zidovudine

RISPERIDONE
- Carbamazepine: Increases metabolism of carbamazepine (reduced plasma concentration).
- Clozapine: With risperidone increases neutropenia.
- Aripiprazole, Clonazepam and Zolpidem: increases side effects such as drowsiness, blurred vision, dry mouth, heat intolerance, flushing, decreased sweating, difficulty urinating, constipation and irregular heartbeat.
- Tricyclic antidepressants can cause increased antimuscarinic side-effects.
- Escitalopram* increases the risk of an irregular heart rhythm that may be serious and potentially life-threatening.

RITONAVIR
- Contraceptives*, Oral: Increases metabolism of estrogens (reduced contraceptive effect)
- Efavirenz: Increased risk of toxicity (monitor liver function tests)

- Fluconazole: Increases plasma concentration
- Indinavir: Increases plasma concentration of indinavir
- Levonorgestrel*: Increases metabolism of levonorgestrel (reduced contraceptive effect)
- Medroxyprogesterone*: Increases metabolism of Medroxy-progesterone
- Methadone: Reduced plasma concentration of methadone
- Norethisterone*: Increases metabolism of norethisterone (reduced contraceptive effect)
- Quinidine*: Increased plasma-quinidine concentration (increased risk of ventricular arrhythmias)
- Saquinavir*: Increases plasma concentration of saquinavir

RITONAVIR+ SAQUANAVIR

- Loperamide, Methylprednisolone Ritonavir may significantly increase the blood levels
- Atorvastatin: Combining these medications may significantly increase the blood levels of atorvastatin.
- *Azithromycin: Combining these medications can increase the risk of an irregular heart rhythm that may be serious and potentially life-threatening.
- Budesonide: Saquinavir may significantly increase the absorption of budesonide into the blood stream.

ROTAVIRUS

- Hydrocortisone: The administration of live, attenuated viral or bacterial vaccines during immunosuppressant or intense antineoplastic therapy may be associated with a risk of disseminated infection due to enhanced replication of vaccine virus or bacteria in the presence of diminished immune competence
- Etanercept, leflunomide, Methotrexate: risk for developing an infection from the vaccine or have a reduced response to the vaccine.

RUBELLA VIRUS

- Betamethasone: The administration of live, attenuated viral or bacterial vaccines during immunosuppressant or intense antineoplastic therapy may be associated with a risk of disseminated infection due to enhanced replication of vaccine virus or bacteria in the presence of diminished immune competence
- Bleomycin, Methotrexate: risk for developing an infection from the vaccine or have a reduced response to the vaccine.

RUBELLA VACCINE

- Betamethasone: The administration of live, attenuated viral or bacterial vaccines during immunosuppressant or intense antineoplastic therapy may be associated with a risk of disseminated infection due to enhanced replication of vaccine virus or bacteria in the presence of diminished immune competence
- Bleomycin, Methotrexate: risk for developing an infection from the vaccine or have a reduced response to the vaccine.

SODIUM VALPROATE: *see* Valproate

SOLUBLE INSULIN: *see* Insulins

SPIRONOLACTONE
- Alcohol, Amlodipine, Atenolol, Chlorpromazine, Diazepam, Fluphenazine, Glyceryl trinitrate, Halothane, Hydralazine, Isosorbide dinitrate, Ketamine, Levodopa, Methyldopa, Nifedipine, Nitrous oxide, Propranolol, Thiopental, Sodium nitroprusside, Timolol, Verapamil: Enhanced hypotensive effect
- Amitriptyline: Increased risk of postural hypotension
- Aspirin: Antagonism of diuretic effect
- Carbamazepine: Increased risk of hyponatraemia
- Cyclosporin*: Increased risk of hyperkalaemia
- Cisplatin: Increased risk of nephrotoxicity and ototoxicity
- Clomipramine: Increased risk of postural hypotension
- Contraceptives, Oral: Antagonism of diuretic effect by estrogens
- Dexamethasone, Hydrocortisone, Prednisolone: Antagonism of diuretic effect
- Digoxin*: Plasma concentration of digoxin increased
- Enalapril*: Enhanced hypotensive effect; increased risk of severe hyperkalaemia (monitor plasma-potassium concentration with low-dose spironolactone in heart failure)
- Ibuprofen: Risk of nephrotoxicity of ibuprofen increased; antagonism of diuretic effect; possibly increased risk of hyperkalaemia
- Lithium*: Reduced lithium excretion (increased plasma-lithium concentration and risk of toxicity)
- Potassium salts: Risk of hyperkalaemia

STAVUDINE
- Didanosine*: Increased risk of adverse effects
- Doxorubicin: Doxorubicin may inhibit effect of stavudine
- Zidovudine*: May inhibit effect of stavudine (avoid concomitant use)

STREPTOMYCIN
- Alcuronium*: Enhanced muscle relaxant effect
- Amphotericin B, Cyclosporin*: Increased risk of nephrotoxicity
- Capreomycin, Vancomycin: Increased risk of nephrotoxicity and ototoxicity
- Cisplatin*: Increased risk of nephrotoxicity and possibly of ototoxicity
- Furosemide*: Increased risk of ototoxicity
- Neostigmine*: Antagonism of effect of neostigmine
- Pyridostigmine*: Antagonism of effect of pyridostigmine
- Suxamethonium*, Vecuronium*: Enhanced muscle relaxant effect

SULFADIAZINE
- Methotrexate: Risk of methotrexate toxicity increased
- Pyrimethamine , Sulfadoxine +Pyrimethamine*: Increased antifolate effect
- Thiopental: Enhanced effects of thiopental
- Warfarin*: Enhanced anticoagulant effect

S.N.	Main list	Complementary list	Dosage form
	Rifampicin		capsule or tablet, 150mg, 300mg, 450 mg
	Streptomycin		powder for injection, 1 g (as sulfate) in vial
	Complementary list: Second-line medicines for the treatment of multi-drug resistant tuberculosis (MDR-TB) – to be made available only in specialised centres adhering to standard treatment protocol.		
		Amoxicillin + Clavulanic acid	tablet, 500 mg + 125 mg, 875 mg+125 mg
		Bedaquiline	tablet, 100mg
		Capreomycin	powder for injection, 1 g in vial
		Clofazimine	capsule, 100 mg
		Cycloserine	capsule, 250 mg
		Ethionamide	tablet, 125, 250 mg
		Kanamycin	powder/solution for injection 1 g in vial/ ampoule
		Levofloxacin*	tablet, 250 mg, 500 mg
		Linezolid	tablet 300 mg
		Moxifloxacin*	tablet, 400 mg
		Ofloxacin*	tablet, 200 mg
		**To be decided based on availability and programme considerations.*	
		P-aminosalicylic acid (PAS)	granules, 4 g, 9.2 g in sachet; tablet 500 mg
		Pyrazinamide	tablet, 150 mg, 400 mg, 500 mg

6.3 Antifungal Medicines